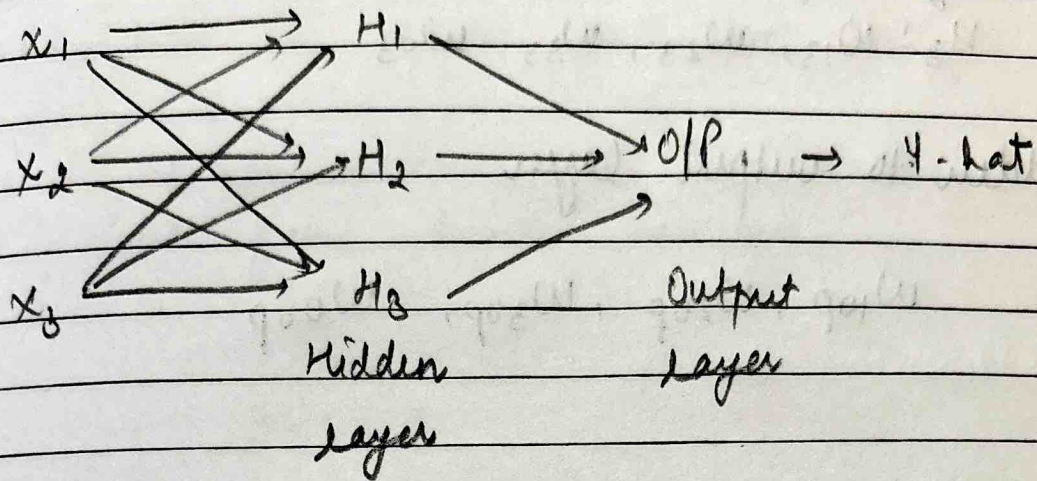


COURSEWORK-1

Problem-1



- i) The network consists of
 Input layer - x_1, x_2, x_3
 Hidden layer - H_1, H_2, H_3
 Output layer - OP

Network is fully connected neural network.

Weights between Input layer and hidden layer

weight is represented as w_{ij}

where i : index of input node
 j : index of hidden node

Thus each hidden neuron receives:

- 3 input weights
- 1 bias



So, for the hidden layers weights will be:

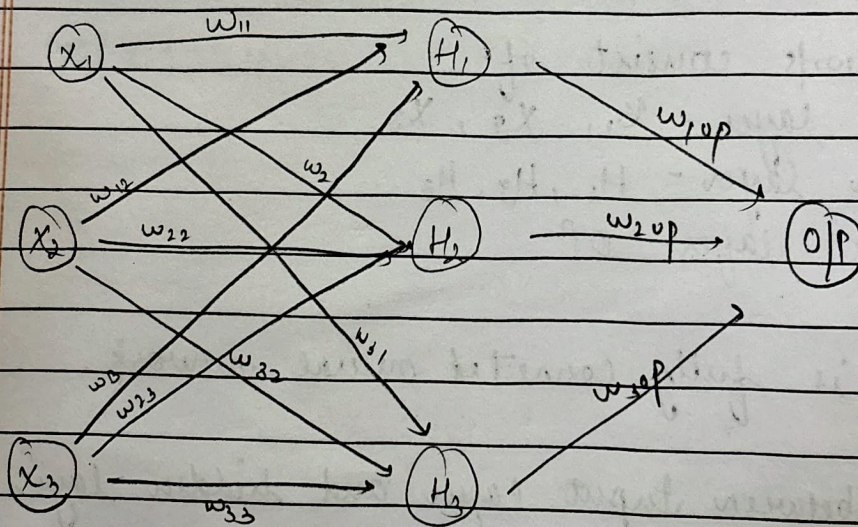
$$H_1: w_{11}, w_{21}, w_{31}, w_{01}$$

$$H_2: w_{12}, w_{22}, w_{32}, w_{02}$$

$$H_3: w_{13}, w_{23}, w_{33}, w_{03}$$

Hidden to output layer

$$w_{1op}, w_{2op}, w_{3op}, w_{0op}$$



ii) Parameters between hidden and output layer

Each hidden neuron receives

3 input weights

1 bias

parameters for one hidden neuron are:

$$3+1=4$$



Therefore, there are 3 hidden neurons:

$$3 \times 4 = 12$$

Parameters b/w hidden and output layer

The O/P neuron receives:

3 weights from hidden neurons

1 bias

$$3 + 1 = 4$$

Total number of parameters

$$4 + 12 = 16$$

iii) Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Hidden neuron H_1

$$Z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + w_{01}$$

$$H_1 = \sigma(Z_1)$$

$$H_1 = \frac{1}{1 + e^{-Z_1}}$$

$$H_1 = \sigma(w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + w_{01})$$



Hidden neuron H_2

$$Z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 + w_{02}$$

$$H_2 = \sigma(Z_2)$$

$$H_2 = \frac{1}{1 + e^{-Z_2}}$$

Hidden neuron H_3

$$Z_3 = w_{13}x_1 + w_{23}x_2 + w_{33}x_3 + w_{03}$$

$$H_3 = \sigma(Z_3)$$

$$H_3 = \frac{1}{1 + e^{-Z_3}}$$

Propagation for output layer

weighted sum

$$Z_{op} = w_{1op}H_1 + w_{2op}H_2 + w_{3op}H_3 + w_{0op}$$

$$y = \sigma(Z_{op})$$

$$y = \frac{1}{1 + e^{-Z_{op}}}$$



(iv) The Mean Squared Error Loss is:

$$L = \frac{1}{2} (y - \hat{y})^2$$

where L = Loss

y = actual output

$\hat{y}$ = predicted output

Output neuron equation is:

$$\hat{y} = \sigma(z_{op})$$

Sigmoid derivation:

$$\sigma'(z) = \sigma(z) (1 - \sigma(z))$$

$$\frac{\partial y}{\partial z_{op}} = y(1-y)$$

Gradient for output layer weights

Derivative of loss

$$\frac{\partial L}{\partial y} = -(y - \hat{y}) = (\hat{y} - y)$$

Derivative of sigmoid

$$\frac{\partial y}{\partial z_{op}} = y(1-y)$$



Derivative for weighted sum

$$\frac{\partial Z_{op}}{\partial w_{2op}} = H_3$$

$$\frac{\partial L}{\partial w_{3op}} = (4-y) \cdot 4 \cdot (1-y) H_3$$

$$\frac{\partial L}{\partial w_{2op}} = (4-y) \cdot 4 \cdot (1-y) H_2$$

$$\frac{\partial L}{\partial w_{1op}} = (4-y) \cdot 4 \cdot (1-y) H_1$$

Gradient of Hidden Neuron

for H_3

Apply chain rule

$$\frac{\partial L}{\partial w_{13}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial Z_{op}} \cdot \frac{\partial Z_{op}}{\partial H_3} \cdot \frac{\partial H_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{13}}$$

Loss Derivative

$$\frac{\partial L}{\partial y} = (4-y)$$

Output sigmoid derivative

$$\frac{\partial y}{\partial Z_{op}} = 4(1-y)$$

Output sum weighted derivative

$$\frac{\partial Z_{op}}{\partial H_3} = w_{3op}$$

Hidden layer sigmoid derivative

$$\frac{\partial H_3}{\partial Z_3} = H_3 (1 - H_3)$$

Hidden weighted sum derivative:

$$\frac{\partial Z_3}{\partial w_{13}} = x_1$$

final

$$\frac{\partial L}{\partial w_{13}} = (y - \hat{y}) \hat{y} (1 - \hat{y}) \cdot w_{3op} H_3 (1 - H_3) x_1$$

Gradient for all hidden layers:

$$\frac{\partial L}{\partial w_{13}} = (y - \hat{y}) \hat{y} (1 - \hat{y}) \cdot w_{3op} H_3 (1 - H_3) x_i$$